

# The Volume-Law Entropy Normalization in the Causal Diamond: What Can and Cannot Be Derived Without Quantum Gravity

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## Abstract

In companion papers, we derived the MOND interpolation function from the competition between area-law and volume-law entropy in the cosmological causal diamond, obtaining  $\mu(x) = kx/(1+kx)$ . The numerical value of  $k$  depends on the volume-law entropy normalization: one choice gives  $k = 1.003$  (0.3% from exact simple MOND), the other gives  $k = 1.34$  (34% off). Here we trace the discrepancy to a single factor of  $4/3$ —the ratio of the sphere volume  $(4\pi/3)r^3$  to  $\pi r^3$ —arising from two equally legitimate definitions of the volume-law entropy density. We show that (a) the functional form of the interpolation function is independent of this choice, (b) the predicted redshift evolution  $a_0(z)/a_0(0)$  is independent of this choice, and (c) resolving the ambiguity requires a microscopic theory of volume-law entropy—a problem at the frontier of quantum gravity. We identify precisely what the causal diamond framework derives and where it reaches the boundary of current theoretical physics.

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## 1. The Ambiguity

In Ref. [1], we derived the MOND interpolation function from the ratio of area-law to total entropy at each cross-section of the cosmological causal diamond. The area-law entropy is the standard Bekenstein-Hawking holographic bound [2]:

$$S_{\text{area}} = A/(4l_{\text{Pl}}^2) = \pi r^2/l_{\text{Pl}}^2 \quad (1)$$

The volume-law entropy, representing dark energy degrees of freedom following Verlinde [3], can be defined in two ways.

### 1.1 Definition A: Redistribute horizon entropy

The de Sitter horizon has entropy  $S_{\text{dS}} = \pi l^2/l_{\text{Pl}}^2$  and encloses volume  $V_{\text{dS}} = (4\pi/3)l^3$ , where  $l$  is the de Sitter radius (or, after the correction of Ref. [4], the physical event horizon  $d_{\text{eh}}$ ). Distributing this entropy uniformly gives a density:

$$\rho_A = S_{\text{dS}} / V_{\text{dS}} = 3 / (4l_{\text{Pl}}^2) \quad (2)$$

The volume-law entropy within a sphere of radius  $r$  is then:

$$S_{\text{vol}}^A = \rho_A \times (4\pi/3)r^3 = \pi r^3 / (l_{\text{Pl}}^2) \quad (3)$$

Note that the factor  $(4\pi/3)$  in the sphere volume cancels against the  $(4\pi/3)$  in the denominator of  $\rho_A$ , leaving  $\pi r^3$  in the numerator.

### 1.2 Definition B: Direct volume scaling

Alternatively, one can write the volume-law entropy as the volume divided by the natural length and area scales of the problem [1]:

$$S_{\text{vol}}^{\text{B}} = V / (l_{\text{Pl}}^2) = (4\pi/3)r^3 / (l_{\text{Pl}}^2) \quad (4)$$

Here the  $(4\pi/3)$  from the sphere volume remains in the numerator. This corresponds to a density  $\rho_{\text{B}} = 1/(l_{\text{Pl}}^2)$ , which is  $(4/3)$  times larger than  $\rho_{\text{A}}$ .

### 1.3 The ratio

The two definitions differ by exactly  $4/3$ :

$$S_{\text{vol}}^{\text{B}} / S_{\text{vol}}^{\text{A}} = 4/3 \quad (5)$$

This single factor determines whether the interpolation parameter is  $k = 1.003$  (Definition B, as adopted in Ref. [1] and corrected in Ref. [4]) or  $k = 1.34$  (Definition A). The  $4/3$  is the geometric factor of a sphere: it is the ratio of the volume  $(4\pi/3)r^3$  to the quantity  $\pi r^3$  that survives after the density cancellation in Definition A.

## 2. What Does Not Depend on the Choice

The following results of the causal diamond program are independent of whether Definition A or B is adopted.

### 2.1 The functional form of the interpolation function

With either definition, the area-law entropy scales as  $r^2$  and the volume-law entropy scales as  $r^3$ . The gravitational coupling  $\mu = S_{\text{area}} / (S_{\text{area}} + S_{\text{vol}})$  is therefore a ratio of a quadratic to a quadratic-plus-cubic, which in the MOND variable  $x = r_w/r$  takes the form  $\mu(x) = kx/(1+kx)$  for some constant  $k$ . The rational form—selecting the simple MOND function  $\mu = x/(1+x)$  over the standard function  $x/\sqrt{1+x^2}$  and the McGaugh function  $1 - \exp(-\sqrt{x})$ —is forced by the power-law competition between  $r^2$  and  $r^3$ , regardless of the prefactors. No normalization choice can produce a square root or an exponential.

### 2.2 The redshift evolution

The predicted evolution  $a_0(z)/a_0(0) = d_{\text{eh}}(0)/d_{\text{eh}}(z)$  [5] is a ratio of event horizons. Both definitions give  $a_0(z) = C/d_{\text{eh}}(z)$  for different constants  $C$ , but the constant cancels in the ratio. The evolution curve is identical for both normalizations.

### 2.3 The acceleration constant

The derivation of  $a_0 = c^2/(2\pi r_w)$  in Ref. [6] comes from the causal diamond waist geometry and the circumference relation, not from the entropy balance. It is entirely independent of the volume-law normalization.

| Result                                                      | Depends on 4/3? | Status      |
|-------------------------------------------------------------|-----------------|-------------|
| $\mu(x) = kx/(1+kx)$ [functional form]                      | No              | Derived     |
| Simple MOND selected over standard/McGaugh                  | No              | Derived     |
| $a_0 = c^2/(2\pi r_w) = 1.21 \times 10^{-10} \text{ m/s}^2$ | No              | Derived     |
| $a_0(z)/a_0(0)$ evolution curve                             | No              | Derived     |
| $k = 1.003$ (Definition B)                                  | Yes             | Conditional |
| $k = 1.34$ (Definition A)                                   | Yes             | Conditional |

**Table 1.** Dependence of causal diamond results on the volume-law entropy normalization. The four primary results are independent of the 4/3 factor. Only the absolute calibration of  $k$  is affected.

### 3. Why the Ambiguity Cannot Be Resolved Without Quantum Gravity

Definition A assumes that the volume-law entropy density equals the de Sitter horizon entropy divided by the enclosed volume: a redistribution of boundary modes into the bulk. Definition B assumes the volume-law entropy density is set by the natural scales of the problem (one factor of the horizon length, one factor of the Planck area) without reference to the horizon entropy.

The two definitions correspond to different physical assumptions about the microscopic origin of volume-law entropy:

**Definition A** treats the bulk entropy as *derived from* the boundary. The de Sitter horizon has  $S_{\text{dS}} = \pi^2 l_{\text{Pl}}^2$  degrees of freedom, and these are distributed throughout the interior. The interior has no independent entropy; it inherits it from the boundary. This is the holographic perspective, in which the boundary is fundamental and the bulk is emergent.

**Definition B** treats the bulk entropy as *independent of* the boundary. The thermal de Sitter vacuum fills the interior with a density set by dimensional analysis: one power of the cosmological length scale and one power of the Planck area. The boundary entropy and the bulk entropy are separate contributions. This is closer to the thermodynamic perspective of Verlinde [3], in which the volume-law term represents a genuinely new contribution that competes with the area law.

Determining which microscopic picture is correct requires a theory that can count degrees of freedom in a gravitating de Sitter region—a problem at the intersection of quantum gravity and de Sitter thermodynamics. This is among the deepest open problems in theoretical physics [7, 8]. The causal diamond framework, which operates at the semiclassical level, cannot resolve it.

We note that this ambiguity is not unique to the present work. Verlinde’s emergent gravity program [3] faces the same normalization question, and published commentary has identified errors of comparable magnitude in his derivation [9]. The problem is structural: any framework that uses volume-law entropy in de Sitter space without a microscopic derivation will encounter this (4/3) ambiguity.

### 4. Can Observations Resolve It?

In principle, a sufficiently precise measurement of the MOND interpolation function in the transition regime ( $x = g/a_0 \approx 0.5\text{--}2$ ) could distinguish  $k = 1.003$  from  $k = 1.34$ . The two predictions differ by 34% in  $k$ , which translates to a ~15% difference in  $\mu(x)$  at  $x = 1$ .

However, current galaxy rotation curve data (e.g., the SPARC sample of 175 galaxies [10]) have scatter of 10–20% in the radial acceleration relation at  $x \approx 1$ , dominated by uncertainties in mass-to-light ratios, distance measurements, and inclination corrections. A 15% shift in  $\mu$  is at the edge of current sensitivity. Next-generation surveys (SKA, JWST) may improve the precision sufficiently to resolve the two values of  $k$ , but this is not guaranteed.

We also note that  $k = 1.003$  (Definition B) is observationally indistinguishable from the simple MOND function  $k = 1$  at any foreseeable precision. If Definition B is correct, the causal diamond framework predicts that the simple function is exact. If Definition A is correct, the framework predicts a measurable 34% departure from the simple function. This is itself a testable prediction: a future measurement ruling out  $k > 1.2$  would favor Definition B; a measurement finding  $k \approx 1.3\text{--}1.4$  would favor Definition A.

### 5. Conclusion

The causal diamond framework for MOND rests on the competition between area-law ( $r^2$ ) and volume-law ( $r^3$ ) entropy. The power-law scaling uniquely determines the functional form of the interpolation function as  $\mu(x) = kx/(1+kx)$ —the simple MOND function—and this result is exact regardless of how the volume-law entropy is normalized.

The absolute calibration of the transition scale (the value of  $k$ ) depends on a single factor of 4/3 between two legitimate definitions of volume-law entropy density. One definition (redistribute horizon entropy) gives  $k = 1.34$ . The other (direct volume scaling) gives  $k = 1.003$  with the event horizon correction [4]. Resolving this requires a microscopic theory of volume-law entropy in de Sitter space, which is a problem at the frontier of quantum gravity.

The following results are derived and do not depend on this resolution:

- The MOND interpolation function is the simple function  $\mu = kx/(1+kx)$ , not the standard or McGaugh function.
- The MOND acceleration constant is  $a_0 = c^2/(2\pi r_w) = 1.21 \times 10^{-10} \text{ m/s}^2$  (1.0% of observed).
- The redshift evolution  $a_0(z)/a_0(0) = d_{\text{eh}}(0)/d_{\text{eh}}(z)$  predicts a 47% increase at  $z = 2$ .

The causal diamond framework derives everything that can be derived at the semiclassical level. The one quantity it cannot fix—the entropy normalization—is the quantity that requires quantum gravity. This is where the framework meets the boundary of known physics.

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## References

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